Naïve Bayes - 2

03 November 2025 20:50

We're talking about **Naive Bayes classifiers**, which use:

$$P(c|x) \propto P(c) \times P(x|c)$$

where x is your feature vector (e.g., the words in an email) and c is the class (like "spam" or "ham").

Different variants of Naive Bayes differ in **how they model** $P(x|c)$ — i.e., what they think the data looks like.

⚙️ 2. Bernoulli Naive Bayes

📖 Idea

Each feature is **binary** — it only matters whether a word is *present* or *absent* in a document.
You're not counting how many times it appears; just whether it shows up at all.

So, for every word w_i and class c:

$$P(w_i = 1|c) = \text{probability that word } w_i \text{ appears in a document of class } c$$

and

$$P(w_i = 0|c) = 1 - P(w_i = 1|c)$$

Vocabulary:

$$V = \{\text{buy}, \text{free}, \text{hello}\}$$

Training data:

Document	Words present	Class
D1	buy, free	spam
D2	buy	spam
D3	hello	ham
D4	free, hello	ham

🔹 Case B: When people say "Binomial Naive Bayes" but actually mean Multinomial Naive Bayes

That's much more common in NLP.
Here, each document is a **bag of words**, and features are **word counts** (not just presence/absence).

$$P(w_i|c) = \frac{\text{count}(w_i, c)}{\sum_j \text{count}(w_j, c)}$$

Class	Document	Words (with counts)
spam	D1	buy(2), free(1)
spam	D2	buy(1)
ham	D3	hello(2)
ham	D4	free(1), hello(1)

Total counts per class:

- spam: buy=3, free=1, hello=0 → total=4
- ham: buy=0, free=1, hello=3 → total=4

Then:

$$P(buy|spam) = 3/4 = 0.75$$
$$P(free|spam) = 1/4 = 0.25$$
$$P(hello|spam) = 0/4 = 0$$

CountVectorizer turns **text documents** (like sentences, paragraphs, or emails) into a **matrix of word counts** — a numeric format that models like Naive Bayes or Logistic Regression can use.

Eg.

```
docs = [
  "free win money",
  "win a free trip",
  "hello friend"
]
```

(Bag of words Algo)

Vocab = ["free", "win", "money", "a", "trip", "hello", "friend"]

(features) f1 f2 f3 f4 f5 f6 f7

free → 0
win → 1
money → 2
a → 3
trip → 4
hello → 5
friend → 6

Document	free	win	money	a	trip	hello	friend
"free win money"	1	1	1	0	0	0	0
"win a free trip"	1	1	0	1	1	0	0
"hello friend"	0	0	0	0	0	1	1

Naive Bayes - 1

→ general classification algo

$$P(y=1|\text{text}) = \frac{P(y=1)\,P(\text{text}|y=1)}{P(\text{text})}$$

$$X = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Multinomial N.B

$$P(\text{spam}|\text{text}) \propto P(y=1)\,P(\text{text}|y=1)$$

$$x = [1\ 1\ 1\ 0\ 0\ 0\ 0]$$

Bernoulli NB

$$P(w_1 \cdots w_d | y=1)$$

(Naive Assumption)

(Naive Assumption)

After NA (C.C.I) →

$P(w_1 \dots w_n | y=1) = P(w_1|y=1) \times P(w_2|y=1) \times \dots$

Problem → Train → $P(y=1)$, $P(y=0)$ & $P(w_i|y=1)$, $P(w_i|y=0)$

Test → $P(y=1|E) = a$

$E =$ ("you won jackpot") → $P(y=0|E) = b$

$\left(\frac{a}{b}\right) > 1 \Rightarrow$ spam else ham

Problem → TR = (20 unique words = Vocab size)

(Test) → " w_1 Pre-regularisation, Forhan-it-all won jackpot "

Likelihood $P(w_1|y=1) = 0$

↳ $P(\text{spam}|\text{text}) = \dots(0) = 0$

$P(w_j|y=1) = \frac{n_{j1}}{n_1}$ L.S → $\frac{n_{j1} + \alpha}{n_1 + V\alpha}$

$\alpha = 1$ (For simplicity) $P(w_j|y=1) = \frac{\alpha}{n_1 + \alpha V} = \frac{1}{(\ )}$

[V → No of unique words in vocab]

Tr Data

Document	Words present	Class
D1	buy, free	spam
D2	buy	spam
D3	hello	ham
D4	free, hello	ham

Without Smoothing $P(\text{buy}|\text{spam}) = \frac{2}{3}$

With Smoothing $\frac{2+\alpha}{3+\alpha 2} = P(\text{buy}|\text{spam})$

Email → (buy free chocolate) → spam

$P(y=1)$ & $P(y=0)$

0.5 0.5

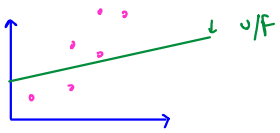
$0.5 * [\ ]$

0.5

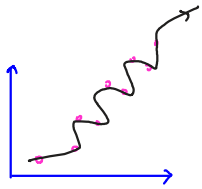
$$\frac{P(y=1 \mid \text{test})}{P(y=0 \mid \text{test})} = \frac{(\quad)}{(\quad)}$$

Effect of α [underfit & overfit]

U/F



U/F



Tr erro = 0
acc = 100%

Bayes theorem =

Effect of α

$$P(y=1 \mid \text{text}) = \frac{\text{Prior } P(y=1) * \text{Likelihoods } P(w_1 \mid y=1) * \cdots P(w_d \mid y=1)}{K}$$

$$P(w_i \mid y=1) = \frac{n_{i1} + \alpha}{n_1 + V\alpha}$$

α = small value ($\alpha=1$) * (overfitting)

($\alpha=10K$) α = v. large value * (Underfitting)

$$\lambda = \frac{c_1 + \alpha}{c_2 + V\alpha} \simeq \frac{\alpha}{\alpha V} \simeq \frac{1}{V}$$

$10^6 + 2 =$

$$P(y=1 \mid \text{text}) = \frac{\text{Posterior}}{\text{Prior } P(y=1) * \frac{1}{V} * \frac{1}{V} * \cdots \frac{1}{V}}$$

$$\frac{P(y=1 \mid text)}{P(y=0 \mid text)} = \frac{}{P(y=0)\ (\frac{1}{V}) * \cdots \frac{1}{V}}$$

$$= \frac{P(y=1)}{P(y=0)} * 1$$

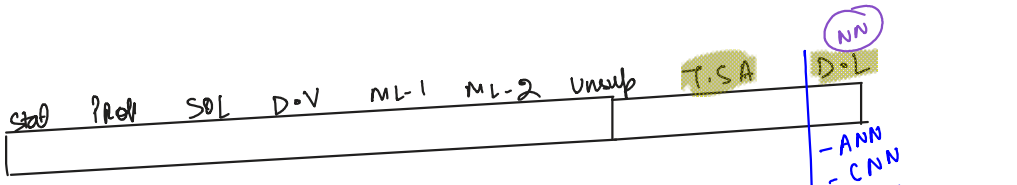
$$P(w_i \mid y=1) = \frac{n_{i1} + \alpha}{n_1 + V\alpha}$$

α = large $= \left(\frac{1}{V}\right) \Rightarrow$ Underfit

Bias — High | Variance → Low

α = small $= \left(\frac{n_{i1}}{n_1}\right) \Rightarrow$ Overfit

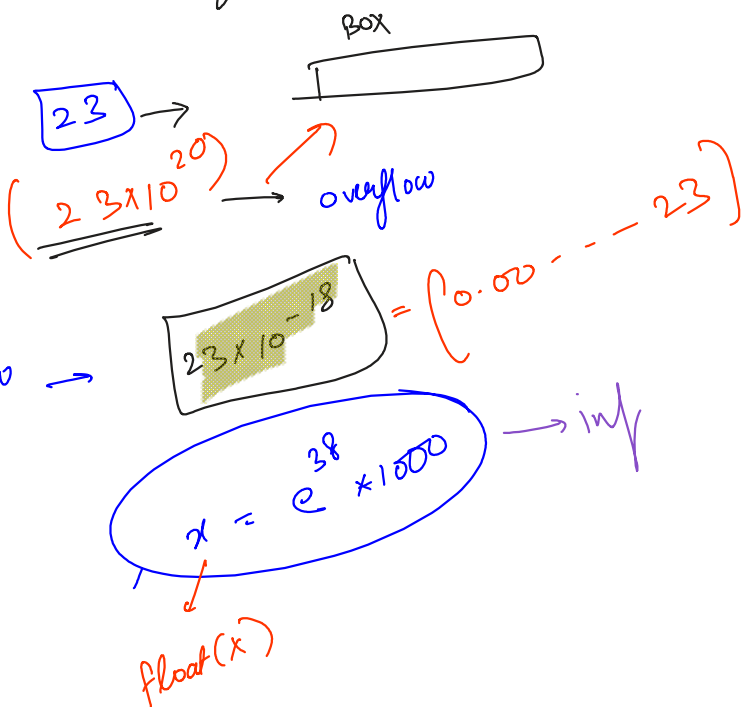
Bias — Low | Variance — high



- ANN
- CNN
- RNN
- LSTMS
- GRU

(Transformer) (Basic NLP)

Underflow & overflow Problem



float(x)

End Goal → E → spam / ham

$$\frac{P(y=1|E)}{P(y=0|E)} = \frac{P(y=1)}{P(y=0)} * \frac{P(w_1|1) * \cdots P(w_d|1)}{P(w_1|0) * \cdots P(w_d|0)}$$

$$\log\left(\quad\right) = \log\left(\frac{P(y=1)}{P(y=0)}\right) + \log\left(\frac{w_1|1}{w_1|0}\right) + \cdots \log\left(\frac{w_d|1}{w_d|0}\right)$$

Feature Importance

E_1 = ("Hello, see you at 5")

E_2 = ("your card is blocked, send otp")

$$P(y=1|E_1) \propto P(y=1) * P(w_1|1)\uparrow\ P(w_2|1)\uparrow * \cdots$$

$$P(y_1=1|E_2) \propto P(y=1)\ P(w_1|1)\uparrow\ P(w_2|1)\uparrow * \cdots$$

"The sky is blue" ↓

$P(\text{blue} \mid \text{The sky is}) \uparrow\uparrow$

"Nigerian Prince"

words → features

(Likelihoods) → weights

Spam → P(Nigerian / spam) → high

Georg Box → (All models are wrong some are useful)

10:16 pm *

Outliers

Train Time → (Rare word → less occurred)

Test Time → (New word (unseen)) → Laplace Smoothing

<k (10)

* (How the data is represented)

$$P(y=1 \mid \text{email}) = \frac{P(y=1)\,P(\text{email} \mid y=1)}{P}$$

Naive Bayes

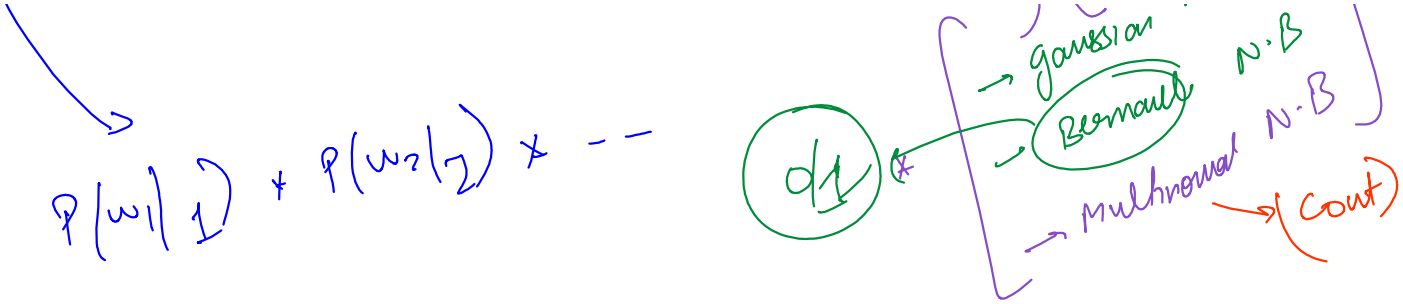
$P(w_1 \mid 1) * P(w_2 \mid 2) * \cdots$

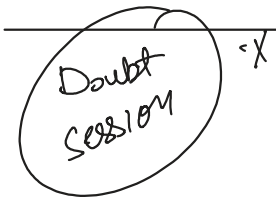
d1 *

→ gaussian N.B

→ Bernoulli N.B

→ Multinomial N.B → (Count)


P(w1|1) + P(w2|2) x - -
d1
gaussian
Bernauli
N.B
Multinomial
N.B
count


Doubt
session